Oracle Analytics Cloud

Private Endpoint Autonomous Database

Connectivity Configuration Guide

Cross-VCN Private Link | Security List Routing | POC-to-Dev Topology
ADB-S Private Endpoint | OAC Private Access Channel | LPG Peering

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1. Executive Summary

This document provides a complete technical reference for configuring Oracle Analytics Cloud (OAC) connectivity to an Autonomous Database Serverless (ADB-S) instance deployed with Private Endpoint Access Only. The ADB-S resides in a POC VCN that is routed exclusively to a Dev VCN containing a standard Oracle Cloud Database (DBCS). No public internet access is configured on the ADB-S.

The configuration requires establishing a Private Access Channel (PAC) from OAC into the POC VCN, configuring Local Peering Gateway (LPG) peering between the POC and Dev VCNs, and ensuring security lists, route tables, and DNS resolution are correctly aligned across all network boundaries.

> **Scope:** This guide assumes OCI tenancy-level administrative access, existing POC and Dev VCN deployments, and a provisioned OAC instance (Enterprise Edition or Professional Edition with PAC support).

2. Network Topology

2.1 Environment Architecture

The deployment spans two OCI Virtual Cloud Networks with strictly controlled inter-VCN routing via Local Peering Gateways. The POC VCN is deliberately isolated from internet egress, making standard OAC public connectivity impossible.

Component	Network	Detail
ADB-S (POC)	POC VCN 10.0.0.0/16	Private Endpoint Only. Subnet: 10.0.1.0/24 (private-adb-sub). No internet gateway.
OAC Instance	OAC Managed VCN	PAC injected into POC VCN private subnet. PAC Subnet: 10.0.2.0/24 (oac-pac-sub).
DBCS (Dev)	Dev VCN 10.1.0.0/16	Standard DB System. Subnet: 10.1.1.0/24 (dev-db-sub). Private access only.
LPG Peering	POC VCN <-> Dev VCN	Bidirectional peering. POC LPG: lpg-poc-to-dev. Dev LPG: lpg-dev-to-poc.

2.2 Traffic Flow Summary

The OAC analytics engine connects through three network hops to reach the private ADB-S:

- OAC managed infrastructure initiates connection through the Private Access Channel
- PAC terminates inside the POC VCN on subnet 10.0.2.0/24
- Traffic routes to ADB-S private endpoint on 10.0.1.0/24 within the same VCN

- ADB-S has a route table entry to Dev VCN via the LPG for any data pipeline traffic
- Return traffic follows symmetric path back through PAC to OAC

> **Critical:** OAC does NOT traverse the LPG to reach the ADB-S. The PAC lands directly in the POC VCN. The LPG exists for ADB-S-to-Dev-DBCS communication only. However, OAC can reach Dev DBCS if routes and security lists are extended.

3. Prerequisites

3.1 OCI Service Requirements

Requirement	Specification
OAC Edition	Enterprise or Professional (PAC requires EE or PE with feature flag)
ADB-S Workload Type	ADW or ATP. Private Endpoint access enabled at provisioning.
OCI Region	All three resources (OAC, ADB-S, DBCS) must be in the same OCI region.
Tenancy Limits	Verify DRG / LPG / VCN limits. At minimum 2 VCNs, 1 LPG pair.
IAM Policies	OAC service must have network access policies for target VCN/subnets.
DNS Configuration	Private DNS resolution for ADB-S private endpoint FQDN.

3.2 Required IAM Policies

The following IAM policy statements must exist in the tenancy (or target compartment) to allow OAC to create and manage Private Access Channels:

```
allow group AnalyticsAdmins to manage analytics-instances in compartment POC
allow group AnalyticsAdmins to manage virtual-network-family in compartment POC

allow service analytics to use virtual-network-family in compartment POC
allow service analytics to manage vnics in compartment POC
allow service analytics to use subnets in compartment POC
allow service analytics to use network-security-groups in compartment POC
```

> **Note:** The 'allow service analytics' statements are mandatory. Without them, PAC creation will fail silently or produce a generic error with no actionable detail.

4. VCN and Subnet Configuration

4.1 POC VCN Structure

Subnet	CIDR	Type	Purpose
private-adb-sub	10.0.1.0/24	Private	ADB-S Private Endpoint VNIC
oac-pac-sub	10.0.2.0/24	Private	OAC Private Access Channel

4.2 Dev VCN Structure

Subnet	CIDR	Type	Purpose
dev-db-sub	10.1.1.0/24	Private	Dev DBCS Instance

4.3 Subnet Requirements for OAC PAC

The OAC Private Access Channel subnet (oac-pac-sub) has specific requirements that must be satisfied before PAC creation:

- Subnet must be private (no public IP assignment)
- Minimum /25 CIDR is recommended; /24 provides safe headroom
- Subnet must have a route table attached (even if initially empty)
- Security list must allow egress to ADB-S subnet on port 1522 (mTLS) or 1521 (TLS)
- The subnet must NOT have a default route to an Internet Gateway
- DHCP Options must include the VCN default DNS resolver (for private endpoint FQDN resolution)

5. Local Peering Gateway Configuration

The Local Peering Gateway (LPG) pair connects the POC VCN to the Dev VCN. This is the sole inter-VCN communication path. No Dynamic Routing Gateway (DRG) or internet transit is involved.

5.1 LPG Provisioning

Step 1. Create LPG in POC VCN named lpg-poc-to-dev in the POC compartment.

Step 2. Create LPG in Dev VCN named lpg-dev-to-poc in the Dev compartment.

Step 3. Establish peering from lpg-poc-to-dev to lpg-dev-to-poc (either direction initiates).

Step 4. Verify peering status shows PEERED on both gateways.

5.2 Route Table Entries

Each VCN requires a route table entry directing traffic to the peer VCN CIDR through the LPG:

POC VCN Route Table (rt-poc-private)

Destination	Target Type	Target
10.1.0.0/16	Local Peering Gateway	lpg-poc-to-dev

Dev VCN Route Table (rt-dev-private)

Destination	Target Type	Target
10.0.0.0/16	Local Peering Gateway	lpg-dev-to-poc

> **Warning:** Do NOT add routes for 0.0.0.0/0 or any internet gateway target in the POC VCN. This would violate the private-endpoint-only posture of the ADB-S.

6. Security List Configuration

Security lists must permit the specific traffic flows required for OAC-to-ADB-S communication and ADB-S-to-Dev-DBCS communication. All rules below follow a least-privilege model.

6.1 POC VCN: OAC PAC Subnet Security List (sl-oac-pac)

Egress Rules

State	Destination	Protocol	Port	Description
Stateful	10.0.1.0/24	TCP	1522	OAC to ADB-S (mTLS)
Stateful	10.0.1.0/24	TCP	443	OAC to ADB-S (HTTPS/REST)

Ingress Rules

State	Source	Protocol	Port	Description
Stateful	10.0.1.0/24	TCP	All	Return traffic from ADB-S

6.2 POC VCN: ADB-S Subnet Security List (sl-adb-private)

Ingress Rules

State	Source	Protocol	Port	Description
Stateful	10.0.2.0/24	TCP	1522	From OAC PAC (mTLS)
Stateful	10.0.2.0/24	TCP	443	From OAC PAC (HTTPS)

Egress Rules

State	Destination	Protocol	Port	Description

Stateful	10.1.1.0/24	TCP	1521	ADB-S to Dev DBCS via LPG
Stateful	10.0.2.0/24	TCP	All	Return traffic to OAC PAC

6.3 Dev VCN: DBCS Subnet Security List (sl-dev-db)

Ingress Rules

State	Source	Protocol	Port	Description
Stateful	10.0.1.0/24	TCP	1521	From ADB-S via LPG

> **Note:** All rules are stateful. Stateful rules automatically permit return traffic, reducing the total rule count. If using Network Security Groups (NSGs) in addition to security lists, ensure NSG rules do not conflict or inadvertently restrict required flows.

7. ADB-S Private Endpoint Configuration

The Autonomous Database must be provisioned with Private Endpoint Access Only. This cannot be changed post-provisioning on a public-endpoint database without re-provisioning.

7.1 Private Endpoint Attributes

Attribute	Value
Access Type	Private Endpoint Access Only
VCN	POC VCN (10.0.0.0/16)
Subnet	private-adb-sub (10.0.1.0/24)
Private IP	Auto-assigned (e.g. 10.0.1.12)
Private Endpoint URL	xxxxxxxx.adb.us-ashburn-1.oraclecloudapps.com
mTLS Required	Yes (default). Port 1522.
NSG (Optional)	nsg-adb-private (if used, must mirror security list rules)

7.2 Allowed VCNs / IP Ranges

Under the ADB-S Network Access control, configure the Access Control List (ACL) to permit inbound connections from the OAC PAC subnet:

```
Access Control Type:  Virtual Cloud Network
VCN:                  POC VCN
IP Notation Type:     CIDR Block
Values:               10.0.2.0/24   (oac-pac-sub)
```

> **Critical:** If the ACL is not configured, the ADB-S private endpoint will reject connections from the OAC PAC subnet even though network-level routing and security lists are correct. This is the most commonly missed step.

8. OAC Private Access Channel

The Private Access Channel (PAC) is the OAC-managed network bridge that injects a VNIC into a customer subnet. This is the mechanism by which OAC reaches private-only resources.

8.1 PAC Creation Procedure

Step 5. Navigate to the OAC instance in the OCI Console.

Step 6. Select Private Access Channel from the Resources menu on the left panel.

Step 7. Click Configure Private Access Channel.

Step 8. Select the POC VCN and the oac-pac-sub subnet.

Step 9. Under DNS Zones, add the private DNS zone for ADB-S: adb.us-ashburn-1.oraclecloudapps.com (adjust for your region).

Step 10. Click Configure. PAC provisioning takes 15–40 minutes.

Step 11. Verify PAC status shows ACTIVE.

8.2 DNS Zone Configuration (Critical)

The DNS zone entry on the PAC is essential. When OAC tries to connect to the ADB-S connection string (e.g., xxxxxxxx.adb.us-ashburn-1.oraclecloudapps.com), it must resolve that hostname to the private IP address of the ADB-S endpoint. The PAC DNS zone configuration tells OAC to use the VCN internal DNS resolver for the specified zone instead of public DNS.

> **Warning:** If the DNS zone is omitted or incorrect, OAC will attempt public DNS resolution, which either fails entirely (no public IP exists) or resolves to a public endpoint that the private-only ADB-S does not expose. The symptom is a connection timeout with no error detail in the OAC UI.

8.3 PAC Egress IP

After PAC activation, note the assigned private IP (visible in the PAC details). This is the source IP from which OAC traffic originates within the POC VCN. Verify this IP falls within the oac-pac-sub CIDR (10.0.2.0/24) for security list rule validation.

9. OAC Database Connection

With the PAC active and network plumbing validated, the OAC connection to ADB-S is configured inside the OAC console.

9.1 Connection Type

Parameter	Value
Connection Type	Oracle Autonomous Data Warehouse / Oracle Autonomous Transaction Processing
Client Credentials	Upload the wallet .zip downloaded from the ADB-S console
Service Name	Select from wallet (e.g., pocadb_medium, pocadb_high)
Username	ADMIN or dedicated analytics schema user
Password	Database user password

9.2 Wallet Considerations

The ADB-S wallet contains TNS entries and certificate material for mTLS authentication. For private endpoint databases, the wallet TNS entries automatically reference the private endpoint hostname. Verify the tnsnames.ora inside the wallet shows the private FQDN, not a public hostname.

```
pocadb_medium = (description=
  (retry_count=20)(retry_delay=3)
  (address=(protocol=tcps)
    (port=1522)
    (host=xxxxxxxx.adb.us-ashburn-1.oraclecloudapps.com))
  (connect_data=(service_name=xxxxxxxx_pocadb_medium.adb.oraclecloud.com))
  (security=(ssl_server_dn_match=yes)))
```

> **Note:** If the wallet was generated before private endpoint was configured, re-download the wallet. Stale wallets may contain public hostnames that will not resolve through the PAC.

10. Validation and Troubleshooting

10.1 Connectivity Validation Checklist

#	Validation Step	Expected Result
1	OAC PAC status in OCI Console	ACTIVE
2	ADB-S status in OCI Console	AVAILABLE
3	LPG peering status (both gateways)	PEERED
4	ADB-S ACL includes 10.0.2.0/24	Entry present
5	Security list sl-oac-pac allows egress to 10.0.1.0/24:1522	Rule present

6	Security list sl-adb-private allows ingress from 10.0.2.0/24:1522	Rule present
7	PAC DNS zone includes adb.<region>.oraclecloudapps.com	Zone listed
8	Wallet tnsnames.ora references private FQDN	Private hostname present
9	OAC connection test using wallet + credentials	Connection Succeeded

10.2 Common Failure Modes

Symptom	Root Cause	Resolution
Connection timeout (no error)	DNS zone missing on PAC	Add ADB-S private DNS zone to PAC config
Connection refused	ADB-S ACL missing OAC subnet	Add 10.0.2.0/24 to ADB-S ACL
ORA-12541: TNS no listener	Security list blocks port 1522	Add egress/ingress rules for 1522
PAC creation fails	Missing IAM service policies	Add 'allow service analytics' policies
ORA-28759: file open failure	Stale wallet or wrong wallet	Re-download wallet from ADB-S
PAC stuck in CREATING	Subnet route table misconfigured	Verify no default IGW route; check subnet associations

10.3 OCI CLI Diagnostic Commands

The following OCI CLI commands can be used to verify configuration state programmatically:

```
# Verify ADB-S private endpoint details
oci db autonomous-database get --autonomous-database-id <ADB_OCID> \
  --query 'data.{"private-endpoint": "private-endpoint", \
    "private-endpoint-ip": "private-endpoint-ip", \
    "subnet-id": "subnet-id"}'

# Verify OAC PAC status
oci analytics analytics-instance get \
  --analytics-instance-id <OAC_OCID> \
  --query 'data."private-access-channels"'

# Verify LPG peering status
oci network local-peering-gateway get \
  --local-peering-gateway-id <LPG_OCID> \
  --query 'data.{"peering-status": "peering-status", \
    "peer-id": "peer-id"}'

# List security list rules for OAC PAC subnet
oci network security-list get \
  --security-list-id <SL_OCID> \
  --query 'data.{"egress": "egress-security-rules", \
    "ingress": "ingress-security-rules"}'
```

11. Extending OAC Access to Dev DBCS (Optional)

If analytics workloads require direct OAC access to the Dev VCN standard DBCS instance (bypassing the ADB-S), additional routing and security list modifications are required to extend the PAC reach across the LPG.

11.1 Additional Route Table Entry

Add a route to the OAC PAC subnet route table (or the rt-poc-private route table if the PAC subnet inherits it):

Destination	Target Type	Target
10.1.1.0/24	Local Peering Gateway	lpg-poc-to-dev

11.2 Additional Security List Rules

sl-oac-pac egress: add TCP/1521 to 10.1.1.0/24 (Dev DBCS listener).

sl-dev-db ingress: add TCP/1521 from 10.0.2.0/24 (OAC PAC subnet).

11.3 Additional PAC DNS Zone

Add the Dev DBCS private DNS zone to the PAC configuration if the DBCS is registered in OCI private DNS. Otherwise, add a custom DNS entry mapping the DBCS FQDN to its private IP.

> **Note:** Cross-VCN PAC connectivity via LPG is supported but adds latency (intra-region, typically sub-millisecond). Monitor query performance in OAC if high-frequency data model refreshes target the Dev DBCS.

12. Security Considerations

- All data in transit is encrypted via mTLS (ADB-S) and TLS (DBCS with Transparent Data Encryption)
- No internet-facing endpoints exist in the POC VCN. Attack surface is limited to LPG-peered traffic
- ADB-S ACL provides database-level network restriction independent of VCN security lists
- OAC PAC creates a managed VNIC; it is not user-accessible via SSH or direct IP
- Wallet rotation should be scheduled and coordinated with OAC connection credential updates
- IAM policies should be scoped to specific compartments, not tenancy-wide, in production

- Network Security Groups (NSGs) are recommended in production for granular per-VNIC control

Appendix A: Complete Network Flow Matrix

Source	Destination	Protocol	Port	Path	Purpose
OAC (PAC)	ADB-S PE	TCP	1522	Intra-VCN	Analytics queries (mTLS)
OAC (PAC)	ADB-S PE	TCP	443	Intra-VCN	REST API / HTTPS
ADB-S PE	Dev DBCS	TCP	1521	LPG	DB Link / data pipeline
OAC (PAC)*	Dev DBCS	TCP	1521	LPG	Direct analytics (optional)

** Requires optional configuration per Section 11.*

End of Document